Seongwon (Gabriel) Yoon

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Research Interests

Optical Interconnects, Silicon Photonics, Network Architecture, AI Infrastructure

Education

Georgia Institute of Technology — Atlanta, GA, USA
Ph.D. in Electrical & Computer Engineering (PI: Prof. Shimeng Yu) — *Aug. 2024 – May. 2029 (expected)*

Seoul National University — Seoul, Korea
B.S. in Materials Science and Engineering, Summa Cum Laude (GPA 4.11/4.3) — *Mar. 2015 – Aug. 2022*

Korea Science Academy of KAIST — Busan, Korea
High School Diploma — *Mar. 2012 – Feb. 2015*

Research Experience

Georgia Institute of Technology — Atlanta, GA, USA
Graduate Research Assistant, Laboratory for Emerging Devices and Circuits — *Aug. 2024 – Present*

- Co-designed collective communication with photonic switch configuration and GPU placement.
- Laid out a silicon photonic MZI switch and simulated it in Lumerical FDTD, MODE, HEAT, INTERCONNECT.
- Extended the htsim packet-level simulator with new GPU fabrics and NVSwitch baselines for MoE workloads.
- Built a full-package Ansys thermal model and energy budget for a co-packaged optical GPU node.

Seoul National University — Seoul, Korea
Undergraduate Researcher, Neuromorphic Materials and Devices Lab — *Mar. 2019 – Oct. 2021, Jun. 2023 – Aug. 2024*

- 1.4M 6T2R phase change memory synaptic arrays with stochastic leaky-integrate-and-fire neurons simulation

Work Experience

Imagoworks Inc. — Seoul, Korea
Software Engineer, Deep CAD Team — *Jul. 2022 – Feb. 2023*

- Fine-tuned generative models for 3D image dental solutions

Samsung Electro-Mechanics — Suwon, Korea
Research Intern, Materials Research Group — *Jan. 2022 – Feb. 2022*

- Analyzed TEM data for MLCC ceramic dielectric

Projects

RRAM-Integrated CMOS Image Sensor Pixel Readout Chip Tape-out — Georgia Tech
180 nm CMOS, 3-metal MPW, National NanoFab Center (NNFC), Korea — *2025*

- Designed binary current-mirror pixels, a shift-register pixel decoder, and a TIA readout for RRAM weights.
- Completed schematic, layout, pad-frame planning, and pre-tape-out simulation.

Publications

- **Seongwon Yoon**, Shimeng Yu, An Optical KV Cache Network for Disaggregated LLM Serving, arXiv preprint arXiv:2610.18444, 2026
- **Seongwon Yoon**, Pin-Jun Chen, Shimeng Yu, Cross-Layer Analysis of Thermal Tuning Stalls in Wafer-Scale Optical Interconnects for LLM MoE Training, arXiv preprint arXiv:2608.24637, 2026
- Faaiq Waqar, Ming-Yen Lee, **Seongwon Yoon**, Seongkwang Lim, Shimeng Yu, CMOS+X: Stacking Memories Based on Oxide Transistors upon GPGPU Platforms, International Symposium on Memory Systems (MEMSYS 2025)
- **Seongwon Yoon**, Janak Sharda, Omkar Phadke, Shimeng Yu, A CMOS Image Sensor Pixel Readout Circuit Design Using Back-End-of-Line (BEOL)-Compatible Oxide Transistors, IEEE International Midwest Symposium on Circuits and Systems (MWSCAS 2025), Best Paper Award

- Yu Gyeong Kang, Masatoshi Ishii, Jaeweon Park, Uicheol Shin, Suyeon Jang, **Seongwon Yoon**, Mingi Kim, Atsuya Okazaki, Megumi Ito, Akiyo Nomura, Kohji Hosokawa, Matthew BrightSky, Sangbum Kim, Solving Max-Cut Problem Using Spiking Boltzmann Machine Based on Neuromorphic Hardware with Phase Change Memory, Advanced Science, 2024
- Jungeun Park, **Seongwon Yoon**, Hannah Kim, Youngjun Kim, Uilyong Lee, Hyungseog Yu, Clinical Validity and Precision of Deep Learning-Based Cone-Beam Computed Tomography Automatic Landmarking Algorithm, Imaging Science in Dentistry, 2024
- **Seongwon Yoon**, Uicheol Shin, Sangbum Kim, Solving Constraint Satisfaction Problem with Spiking Neural Network Based on 1.4M 6T2R PCM Synaptic Array with 1.6K Stochastic LIF Neurons Neuromorphic Hardware, Korean Conference on Semiconductors (KCS 2021)

Talks

Optical Interconnect for LLM, MoE and Agentic AI — Atlanta, GA, USA
Future of CMOS and Talent Workshop, Memory Wall and Interconnect Wall, Georgia Tech — *Mar. 2026*

Technical Skills

Photonic Simulation: Ansys Lumerical (FDTD, MODE, HEAT, INTERCONNECT)

Device & Multiphysics: TCAD, Ansys Mechanical (thermal and structural)

AI System Modeling: htsim, MixNet, FlexFlow, SimAI, vLLM

Developer Tools: NVIDIA Nsight, Git, LaTeX, Docker, Wireshark

Selected Honors and Awards

Jung-hun Foundation Scholarship — South Korea
Full tuition support for 2 years — *2018 – 2020*

Eminence Scholarship, Seoul National University — South Korea
Full tuition support for 1.5 years — *2015 – 2016*